Designing AI-Augmented Marketing Education: The RFM Educational Design Model as a Human-Centered E-Business Information System*

Tian Vane-Ing [1], William Yeoh[1] and Pang Wai-Man[2]
[1] Hong Kong Metropolitan University, Hong Kong
{vitian, wyeoh}@hkmu.edu.hk
[2] Hong Kong Polytechnic University, Hong Kong
waiman.pang@polyu.edu.hk

Abstract.

As artificial intelligence (AI) continues to transform e-business practices, educational institutions face mounting pressure to equip students with AI competencies for the digital economy. This paper reports an exploratory, first-cycle design science study of the RFM (Reasonable Challenge, Fun, Motivation) education design model, a theoretically grounded framework for designing AI-enhanced educational e-business information systems. Adopting a design science approach, we integrate generative AI tools (ChatGPT, DeepAI, PixVerse), a chatbot platform (Rochat), and a collaborative system (Padlet) into a postgraduate marketing course and monitor the resulting AI-augmented learning outcomes. The findings highlight critical design considerations, including tool accessibility, scaffolded prompt engineering, peer learning dynamics, and the integration of ethical AI discourse. This study contributes a novel motivational design framework for AI-rich business education, together with practical insights for institutions seeking to evolve their educational information systems to meet AI-era skills demands.

1. Introduction

The rapid evolution of artificial intelligence is fundamentally reshaping the e-business landscape, creating an urgent demand for professionals equipped with AI competencies [1][3][25]. Educational institutions must evolve to prepare students for AI-driven marketing environments in which generative models create content, chatbots manage customer relationships, and data-driven insights guide strategic decisions [7][11][17][27][32].

Yet significant gaps persist between AI's transformative potential and its integration into business curricula. Many programs struggle to move beyond conceptual AI discussions to hands-on competency development. Students graduating with traditional business education often lack practical experience with generative AI tools, leaving them unprepared for workplaces increasingly reliant on AI-augmented workflows. [2][4][5][34]

This paper addresses this gap through a design science approach [14], presenting the development and evaluation of an AI-enhanced educational e-business information system guided by the RFM (Reasonable Challenge, Fun, Motivation) educational design model. The RFM education design framework—rooted in established motivational theories including Self-Determination Theory[19] [29][30], Expectancy Theory

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. 6, July 8-10, 2026, Seoul, South Korea.
© The copyright of this paper remains with the author(s).

[24], and Behavioral Learning Theory[12][18]—provides structured design principles for creating engaging, effective AI-rich learning environments.

Distinct from the traditional RFM (Recency–Frequency–Monetary) model widely used in marketing analytics, the RFM Educational Design Model is an original pedagogical framework developed by the course instructor to address the motivational and design challenges of AI-augmented education. While the marketing RFM model is an analytical tool for measuring customer value, the RFM Educational Design Model is a human-centred design framework that specifies how educational information systems should structure tasks, interactions, and feedback to keep learners appropriately challenged, engaged, and motivated.

The paper makes three primary contributions. First, it introduces the RFM Educational Design Model as a theoretically grounded, practically validated framework for designing AI-augmented educational EBIS. Second, it demonstrates how generative AI tools, chatbot platforms, and collaborative systems can be integrated into postgraduate marketing education through a design science approach. Third, it provides evidence-based insights into how students experience challenges, enjoyment, and motivation when learning with AI, contributing to the growing body of knowledge on human-centred EBIS design and AI competency development in business education.

2. Theoretical Foundations and the RFM Educational Design Model

2.1 Theoretical Foundations

The RFM Educational Design Model did not originate in theory alone. It emerged from years of direct pedagogical observation across diverse classroom settings—undergraduate, postgraduate, and distance learning environments—where three recurring patterns became unmistakably clear: when students perceive tasks as excessively challenging, they disengage and give up; when learning feels tedious or boring, participation becomes superficial or performative; and when motivation is absent, students do only the minimum required to pass. These three honest observations—about challenge, enjoyment, and motivation—formed the practical seed from which the RFM Educational Design Model grew.

The design of effective AI-augmented educational environments, however, requires more than practitioner insight; it demands a principled theoretical understanding of why these patterns occur and how to address them systematically through intentional system design. The RFM Educational Design Model is therefore grounded in five established theoretical perspectives drawn from educational psychology and motivational research, each contributing a distinct and complementary lens to the model's design principles. Rather than adopting a single theoretical framework, the RFM model deliberately integrates multiple perspectives to capture the multidimensional nature of student motivation in AI-rich learning contexts—where novelty, uncertainty, and rapid skill acquisition simultaneously create opportunities for deep engagement and risks of disengagement. Together, these five perspectives provide a robust and coherent conceptual basis for the design of AI-augmented educational EBIS that are not only technically functional but also motivationally effective and human-centred. Effective educational information systems must address not only the cognitive but also the motivational dimensions of learning [26]. Five theoretical perspectives inform the RFM model's design principles:

Self-Determination Theory (SDT) [29][30] emphasizes three fundamental psychological needs: autonomy (sense of volition), competence (sense of mastery), and relatedness (sense of connection). Educational systems supporting these needs enhance intrinsic motivation and well-being. SDT provides a foundational rationale for all three RFM components.

Expectancy Theory [24] suggests that motivation depends on the perceived likelihood of success and the value placed on achievement. Students engage more deeply when tasks are challenging yet attainable—a principle central to "Reasonable Challenge." Recent applications in generative AI contexts [10] show that expectancy theory helps explain professionals' adoption of AI tools when they perceive that effort will yield meaningful outcomes.

Attribution Theory[15][16] 21] [33] highlights how beliefs about responsibility for outcomes affect motivation. When students attribute success to effort and skill rather than external factors, they demonstrate greater persistence—supporting both "Reasonable Challenge" and "Motivation" design.

Maslow's Hierarchy of Needs[6][9][23] underscores progression from basic needs to self-esteem and self-actualization. Educational systems addressing esteem needs (confidence, achievement) and self-actualization (creativity, growth) align with RFM's "Motivation" component.

Behavioral Learning Theory and gamification research[12][18][19] [31] demonstrate that activities balancing challenge with skill, incorporating game elements, and providing immediate feedback sustain engagement—principles operationalized in RFM's "Fun" component.

2.2 The RFM Educational Design Model

Building on these theoretical foundations, the RFM educational design model comprises three integrated design dimensions:

I. **Reasonable Challenge**: Designing tasks that balance difficulty and attainability, ensuring students remain engaged without feeling overwhelmed. This operationalizes expectancy theory by maintaining perceived likelihood of success while requiring genuine effort. Tasks should be scaffolded—starting with guided demonstrations, progressing to supported practice, culminating in independent application.

II. **Fun**: Creating learning experiences that are **interesting** (novel, relevant), **interactive** (hands-on, participatory), and **innovative** (cutting-edge, creative)—the "three I's" sustaining curiosity and participation. This dimension operationalizes behavioral learning theory and gamification principles, incorporating elements of play, exploration, and aesthetic engagement.

III. **Motivation**: Inspiring students through three key drivers—the "Three P's":

- *Perceived Usefulness*: Helping students recognize real-world relevance and practical value of learned skills, addressing "why does this matter for my career?"
- *Peer Showcase*: Fostering collaboration and healthy competition by encouraging students to share and celebrate achievements, supporting SDT's relatedness need.
- *Perceived Competence*: Building confidence in abilities and reinforcing a sense of mastery, directly supporting SDT's competence need and addressing self-esteem in Maslow's hierarchy.

The RFM model distinguishes itself from generic motivational frameworks through its explicit integration of these five theoretical perspectives and its operationalization specifically for AI-rich business education contexts. While Self-Determination Theory identifies overarching needs (autonomy, competence, relatedness), RFM translates them into concrete educational system design principles applicable to AI tool integration, assignment structuring, and peer learning orchestration. Figure 1 depicts the conceptual illustration of the RFM Educational Design Model.

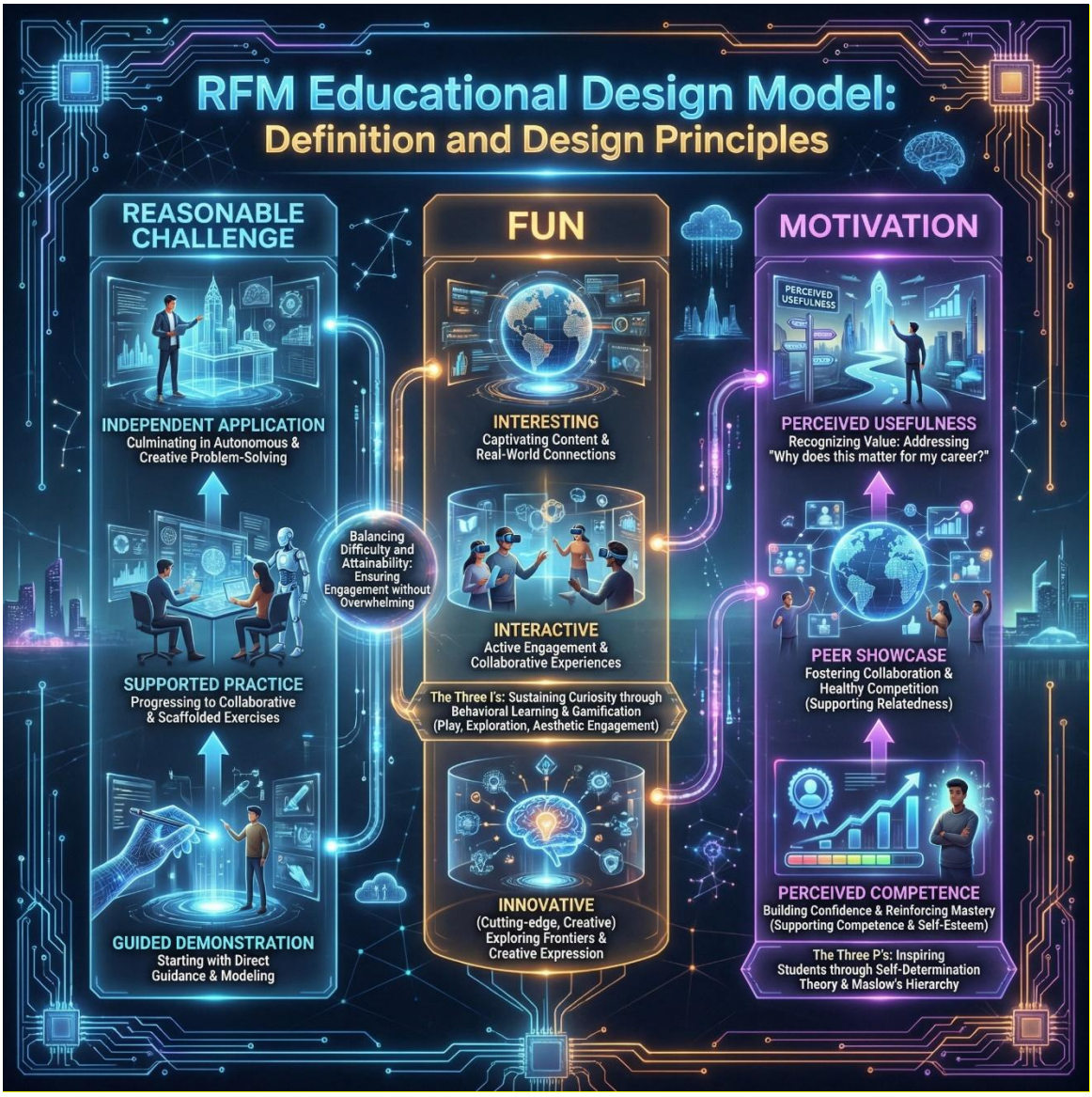


Figure 1. Conceptual Illustration of the RFM Educational Design Model

In short, RFM answers a question that these influential frameworks do not explicitly address: *how do we design AI-augmented educational information systems that are not only technically functional but also motivationally effective for business students with diverse backgrounds, varying technical confidence, and rapidly evolving professional requirements?* It is this domain-specific, practically grounded extension of established motivational theory that constitutes the distinctive contribution of the RFM Educational Design Model. Table 1 summarises the comparative positioning of the RFM Educational Design Model against four established motivational frameworks across key theoretical and design dimensions.

	ARCS Model [8]	**Flow Theory [28]**	**UDL [20]**	**Malone & Lepper [22]**	**RFM Model (This Study)**
Primary Focus	Instructional motivation design	Optimal psychological experience	Inclusive learning design	Intrinsic motivation taxonomy	Competency development in education
Theoretical Type	Prescriptive design model	Descriptive psychological theory	Architectural design philosophy	Descriptive motivational taxonomy	Prescriptive domain-specific design model
Core Components	Attention, Relevance, Confidence, Satisfaction	Clear goals, Immediate feedback, Challenge-skill balance	Engagement (Why), Representation (What), Action & Expression (How)	Individual: Challenge, Curiosity, Control, Fantasy; Interpersonal: Cooperation, Competition, Recognition	Reasonable Challenge, Fun, Motivation
Feedback Mechanism	Implicit confidence building through satisfaction	Immediate feedback as flow condition (descriptive)	Multiple means of action and expression	Performance feedback under Challenge; Recognition under interpersonal	Structured iterative feedback cycles explicit and embedded
Mastery / Competency Link	Not explicitly specified	Implicit challenge-skill balance	Indirect through multiple expression pathways	Not specified motivational categories only	Explicit mastery checkpoints at each scaffolding stage
Interpersonal Dimension	Not addressed	Not addressed	Cooperative expression pathways (indirect)	Explicitly named: Cooperation, Competition, Recognition	Endogenous peer review; collaborative showcase
Endogenous vs. Exogenous	Not distinguished	Intrinsic by nature but not operationalised	Not distinguished	Explicitly distinguished endogenous preferred over exogenous	Endogenous motivational mechanics embedded in task structure
AI Integration	Limited; recent extensions emerging	Emerging AI as flow-enabling scaffold [28]	Strong conceptual alignment domain-general [20]	Not addressed in original taxonomy	Embedded and central GenAI tools are core design elements
Learner Role	Passive recipient of motivationally designed instruction	Self-directed experiencer of optimal states	Active participant across multiple expression pathways	Intrinsically motivated individual and social actor	Active co-creator of AI-generated marketing artefacts
Instructor Role	Motivational designer	Flow condition facilitator	Inclusive architect	Not specified	Scaffolding curator and mastery validator

Operational	Moderate design checklist for instructors	Low abstract psychological state; hard to operationalise directly	Moderate broad principles requiring contextual translation	Low descriptive categories without implementation guidance	High concrete three-stage scaffolding with specific tool and task recommendations

Table 1: Comparative Analysis of RFM and Four Related Motivational Frameworks

3. Course Design: The AI-Augmented EBIS Artifact

This research follows the design science paradigm in information systems[10][14], which seeks to create and evaluate artifacts that solve identified organizational problems. Design science emphasizes both artifact creation (design) and empirical evaluation (science), producing prescriptive knowledge about what designs work and why.

Following design science research principles, the postgraduate marketing course was conceptualised as an AI-augmented educational e-business information system (EBIS)—a socio-technical artifact comprising tasks, AI tools, platforms, interaction flows, and feedback mechanisms, all guided by the RFM Educational Design Model. The course is a postgraduate contemporary marketing course designed to build students' practical AI competencies for e-business marketing environments.

The AI tools and platforms integrated into the course were selected to mirror the generative AI and conversational AI technologies increasingly deployed in real-world e-business practice, ensuring that students' learning experiences were directly relevant to professional marketing contexts. Table 2 summarises the mapping between RFM principles, EBIS design decisions, and the tools deployed.

RFM Principle	Design Decision	Tool/Platform
Reasonable Challenge	Levelled AI tasks; scaffolded instruction	ChatGPT, DeepAI, PixVerse
Fun (Interesting)	Novel AI content creation tasks	ChatGPT, DeepAI, PixVerse
Fun (Interactive)	Gamified chatbot interaction; live demo	Rochat, Dr. Tian chatbot
Fun (Innovative)	Generative visual marketing content	ChatGPT, DeepAI, PixVerse
Motivation (Peer Showcase)	Public gallery-style submission	Padlet
Motivation (Perceived Usefulness)	Authentic real-world marketing tasks	All tools
Motivation (Perceived Competence)	Peer feedback; scaffolded success	Padlet, OLE polls

Table 2. RFM Educational Design Principles, EBIS Design Decisions, and Integrated Tools

3.1 Designing for Reasonable Challenge

The Reasonable Challenge principle guided the design of AI-based assignments that were deliberately calibrated to be attainable yet intellectually stretching. Drawing on Expectancy Theory and Attribution Theory, tasks were structured so that students could achieve meaningful outcomes through effort and experimentation, without requiring prior technical expertise in AI. The principle was operationalised through two complementary assignment types: an individual academic engagement assignment and a series of progressive hands-on AI assignments.

3.1.1 Individual Assignment: Academic Engagement with Generative AI Research

To develop students' critical analytical competencies alongside their practical AI skills, students were assigned an individual academic engagement task in which they analysed a recently published research paper on generative AI in marketing. Students selected two topics from the paper's future research agenda and developed evidence-based arguments drawing on academic databases, secondary data, and real-world marketing examples.

This assignment was carefully designed to balance challenge with support. The challenge dimension required postgraduate-level critical analysis, academic database navigation, and evidence synthesis—skills that stretched students beyond passive consumption of AI concepts. The support dimension provided clear scaffolding: a specific paper was provided, topic choices were clearly delineated, and library database access was available. This balance ensured that students perceived the task as demanding but achievable, consistent with Expectancy Theory's prescription that motivation is maximised when success is attainable through effort.

3.1.2 Hands-On AI Assignments: Progressive Scaffolding

The hands-on AI assignments followed a deliberate three-stage scaffolding structure that progressively built students' competence before requiring independent application:

Demonstration stage: The instructor demonstrated each tool's basic functions in class using prepared examples—for instance, creating perfume advertisement imagery using DeepAI and PixVerse, and building the Dr. Tian AI chatbot using Rochat. These demonstrations provided a concrete reference point and reduced initial anxiety by making the tools feel approachable and achievable.

Guided experimentation stage: Students engaged in low-stakes, in-class activities before any graded work was required. For example, students interacted with the Dr. Tian chatbot by asking course-related questions, discussing prompt effectiveness, and observing how different inputs produced different outputs. This stage built foundational familiarity and confidence in a risk-free environment.

Independent application stage: Having developed foundational competence through the first two stages, students completed graded assignments requiring them to create original marketing content—including AI-generated product designs and virtual brand spokespeople—using the tools they had explored in class.

This progressive scaffolding structure ensured that challenge levels were continuously aligned with students' evolving competence, consistent with SDT's emphasis on competence-supportive environments and with flow theory's prescription for maintaining engagement through optimal challenge-skill balance. Rather than expecting immediate mastery of unfamiliar AI tools, the three-stage structure allowed students to build confidence incrementally, transforming initial apprehension into a sense of growing capability that carried forward into subsequent assignments.

3.2 Designing for Fun: Interactive, Innovative, and Interesting Learning Experiences

The second principle of the RFM Educational Design Model operationalises Fun through three dimensions: Interesting (novel and relevant), Interactive (hands-on and participatory), and Innovative (cutting-edge and

creative). Within the AI-augmented EBIS, Fun was instantiated through three concrete design decisions: gamified chatbot interaction, hands-on generative content creation, and a visual peer showcase platform.

3.2.1 Gamified Chatbot Interaction

Recognising that building an AI chatbot might initially seem technically daunting to postgraduate students, the course instructor first introduced chatbot technology through a low-stakes, playful in-class activity. The instructor developed Dr. Tian, a course-specific AI chatbot built on the Rochat platform, designed with an empathetic and creative personality to respond to students' questions about assignments, marketing concepts, and course logistics. During class, students were invited to interact with Dr. Tian, discovering its patient and contextually appropriate responses in real time.

Figure 2 shows the screenshot of the Dr. Tian AI Chatbot interface built on Rochat.

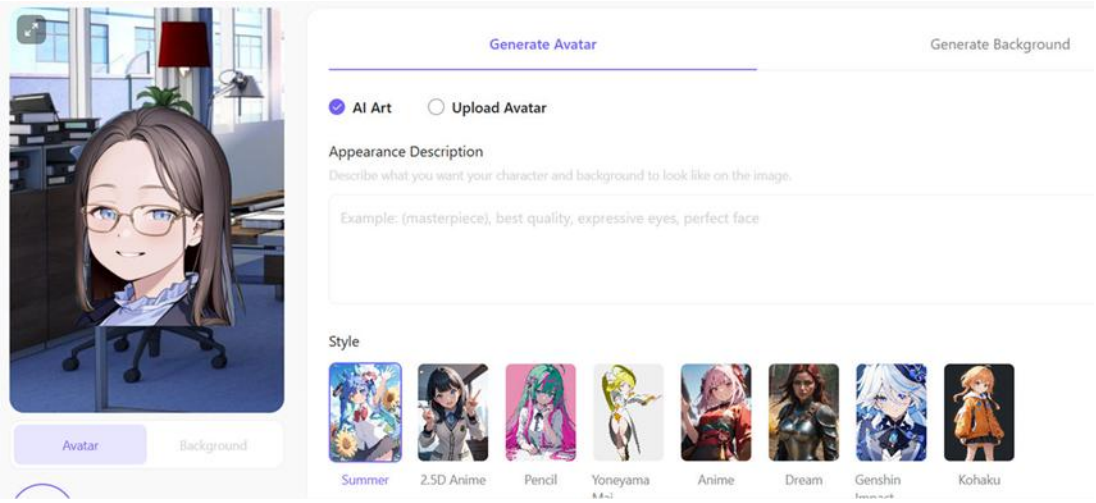


Figure 2. Screenshot of the Dr. Tian AI Chatbot Interface Built on Rochat

Rochat can generate AI art to create avatars based on an uploaded photo or a description of the desired appearance. Additionally, it offers an intriguing feature that allows users to customize the chatbot's personality and select the Large Language Models (LLMs) to be used, as outlined in Figure 3.

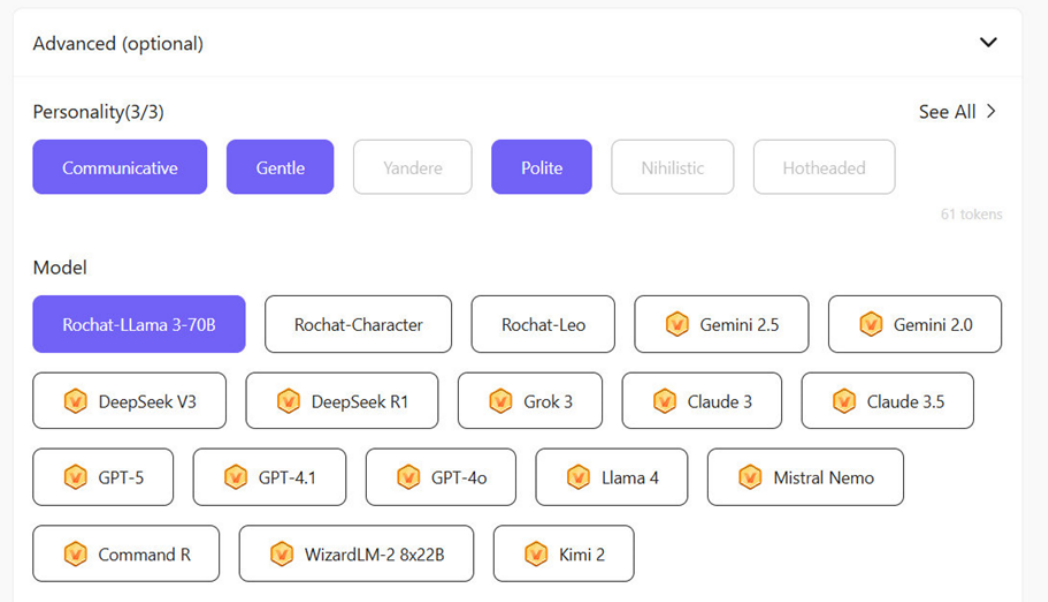


Figure 3. Rochat Avatar Generation and Chatbot Personality Customization Options

This gamified entry point served multiple EBIS design functions. First, it reduced anxiety about the subsequent chatbot-building assignment by providing an enjoyable first-hand demonstration that AI chatbot development is approachable. Second, it illustrated the practical marketing application of conversational AI—namely, how chatbots can manage customer relationships, deliver personalized responses, and sustain brand engagement—thereby directly connecting course content to contemporary e-business practice. Third, it sparked curiosity and intrinsic motivation, consistent with SDT's emphasis on competence-supportive environments and with behavioral learning research showing that gamified activities sustain engagement and flow.

Figure 4 illustrates the in-class guided experimentation stage, capturing students' first direct interaction with the Dr. Tian chatbot. During this session, students were invited to pose course-related questions freely and discover the chatbot's empathetic, contextually appropriate responses in real time. The relaxed, exploratory nature of this activity was deliberately designed to lower the perceived barrier to chatbot technology before any graded work was required.

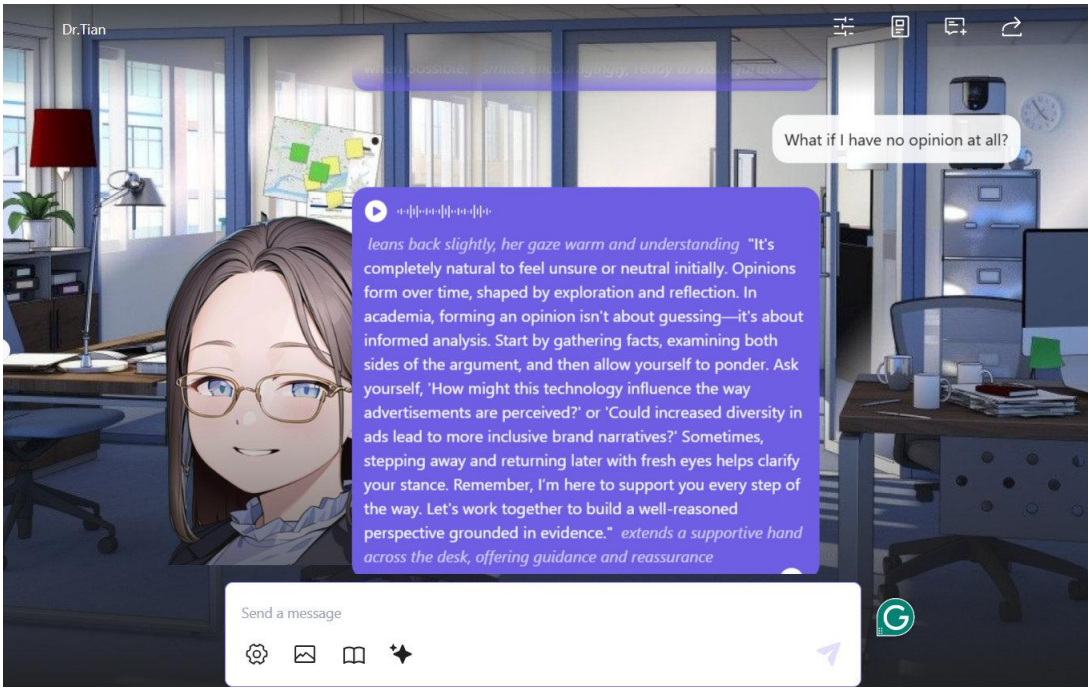


Figure 4. In-Class Demonstration of the Dr. Tian Chatbot During the Guided Experimentation Stage

Figure 5 presents selected examples of student prompts and Dr. Tian's responses during the guided experimentation session. These exchanges illustrate the chatbot's capacity to deliver patient, detailed, and empathetic guidance on assignment-related queries — demonstrating to students that conversational AI can function as a responsive, supportive learning companion as well as a practical marketing tool for managing customer interactions.

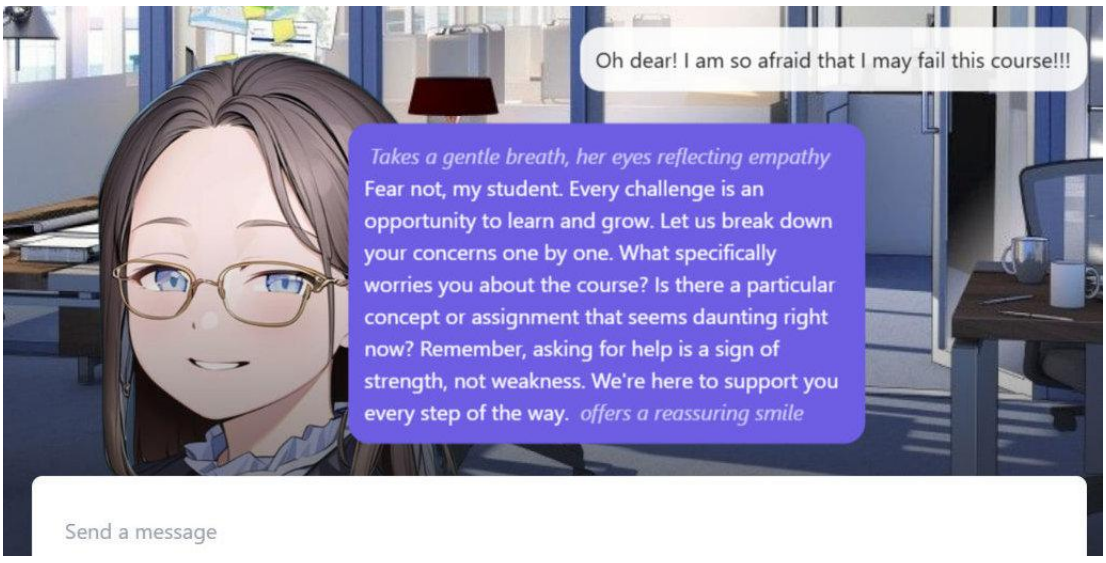


Figure 5. Student Interaction with the Dr. Tian Chatbot: Sample Prompts and Responses

The Rochat platform provided rich functionality appropriate for marketing education: AI-generated avatars based on uploaded photos or text descriptions, customizable chatbot personalities, selectable Large Language Models (LLMs), and voice selection. These features allowed students to experiment with the same design decisions that real e-business practitioners face when deploying AI-powered virtual brand agents. To further demystify chatbot development, the instructor conducted a live demonstration during a tutorial session, building a new chatbot in real time using student inputs, reinforcing the participatory and innovative dimensions of Fun.

Figure 6 documents this live tutorial demonstration, showing the instructor building a new chatbot from scratch using suggestions and input from students in the room. This transparent, participatory exercise made the chatbot development process visible and immediately comprehensible, transforming what might have seemed a complex technical undertaking into an accessible, collaborative creative activity — directly operationalizing the Reasonable Challenge principle by making success feel attainable before students attempted the task independently.

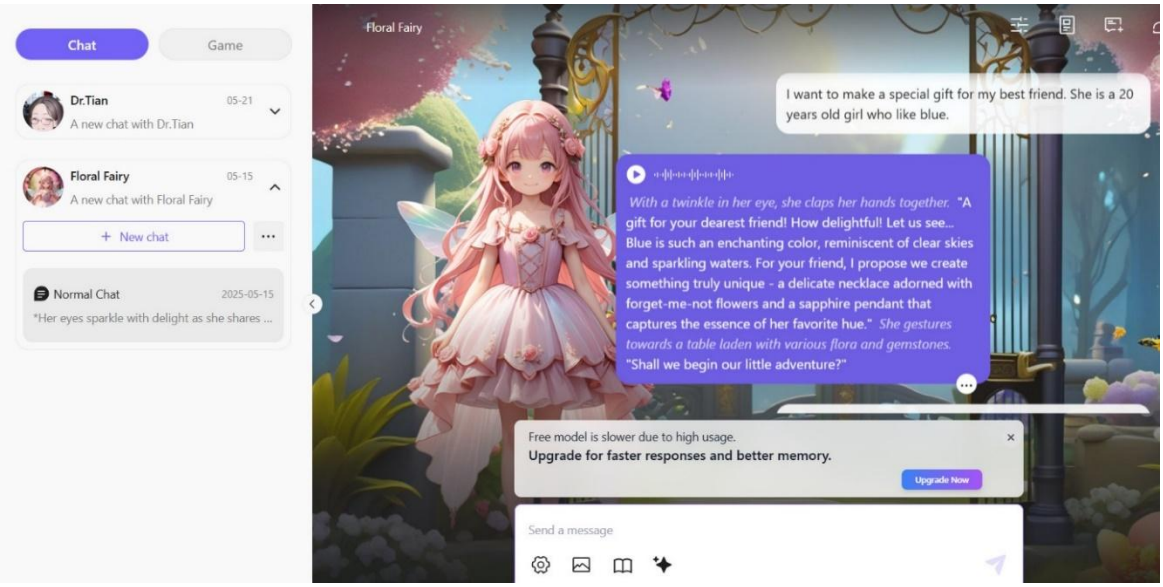


Figure 6. Live Chatbot-Building Demonstration Using Student Inputs in Tutorial Session

3.2.2 Hands-On Generative Content Creation

Rather than passively consuming AI concepts, students actively generated visual marketing content as part of their assessed work. Using tools including ChatGPT, DeepAI, and PixVerse, students created product images, advertisement designs, and AI-generated video content by transforming static images into short animated sequences. These creative, experimental activities directly aligned with the innovative and interactive dimensions of Fun in the RFM Educational Design Model.

This design choice reflects a deliberate departure from lecture-only AI instruction. Consistent with design science principles, the course was treated as an artifact whose components—tasks, tools, and interaction flows—were selected to produce measurable learning outcomes. Hands-on generative content creation positioned students as active producers of e-business-relevant AI outputs rather than passive recipients of conceptual knowledge, directly addressing the curriculum–skills gap identified in the literature.

Figure 7 presents examples of AI-generated marketing visuals produced using ChatGPT, drawn from instructor-prepared handouts distributed to students at the beginning of the course. These instructor-created

examples served a dual pedagogical purpose: first, to demonstrate the creative and commercial potential of generative AI tools before students attempted their own designs; and second, to establish a visible quality benchmark that was achievable yet inspiring — embodying the Reasonable Challenge principle by showing students what was possible through well-crafted prompting without requiring any prior design or technical expertise.



Figure 7. Examples of Ai-Generated Marketing Visuals Using ChatGPT in handouts.

3.2.3 Visual Peer Showcase via Padlet

The third Fun-oriented design element was the deployment of Padlet as a visual, gallery-style showcase platform. Rather than submitting assignments as private instructor–student exchanges, students posted their AI-generated marketing visuals to a shared Padlet board, transforming the assessment process into a public creative exhibition that simultaneously supported peer learning and healthy competition.

The instructor opened the showcase by sharing her own design examples, setting a creative standard and demonstrating the platform's potential. Initially hesitant, students progressively engaged as peers began to share increasingly creative work. Students commented on each other's work, experienced social comparison that stimulated further creative effort, and reported increased confidence in applying AI tools to future marketing tasks. From an EBIS design perspective, Padlet functioned as the collaboration and knowledge-sharing layer of the AI-augmented system, consistent with research on peer showcase as a driver of perceived usefulness and perceived competence.

Figure 8 shows the Padlet showcase board at a representative point in the course, highlighting the diversity and creativity of students' AI-generated marketing designs. The gallery-style layout made individual contributions immediately visible to the entire class, creating a shared creative space that was both motivating and instructive — students could observe a wide range of prompting strategies, design styles, and brand interpretations, learning as much from each other's outputs as from formal instruction. The visible accumulation of high-quality peer work over the semester served as tangible, ongoing evidence of what the cohort was collectively capable of achieving — a powerful reinforcement of both the Fun and Motivation dimensions of the RFM Educational Design Model.

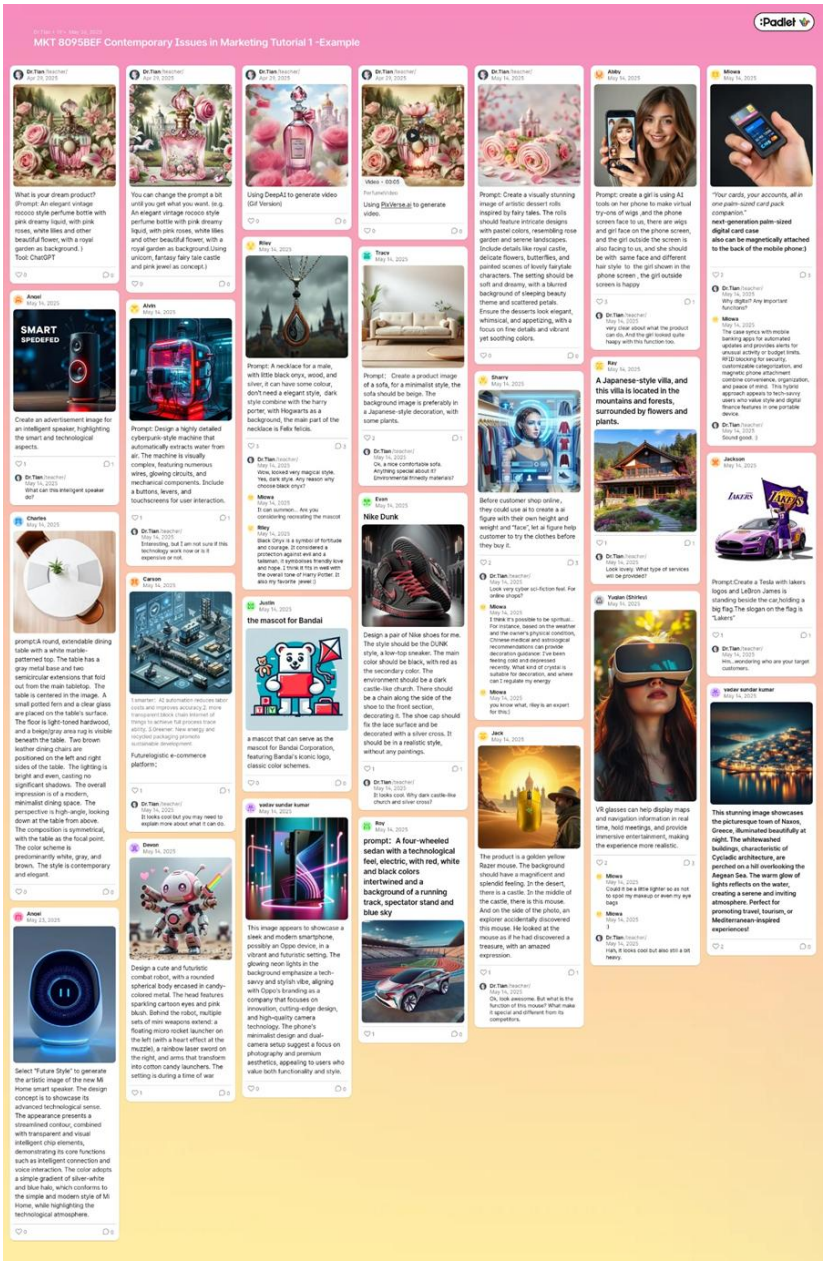


Figure 8. Student AI-Generated Marketing Designs Displayed on the Padlet Showcase Board

3.3 Designing for Motivation: The Three P's

The third RFM principle—Motivation—was operationalised through the Three P's: Perceived Usefulness, Peer Showcase, and Perceived Competence. Together, these three drivers ensured that students not only engaged with AI tools during class but also developed a lasting sense of their value and relevance for professional practice.

Perceived Usefulness was cultivated by embedding all AI tasks within authentic marketing scenarios drawn from real e-business practice. Students were not asked to experiment with AI in the abstract; instead, they used generative AI to design marketing visuals for actual product concepts, built AI chatbots that functioned

as virtual brand spokespeople, and reflected critically on the implications of AI-generated content for marketing ethics, intellectual property, and consumer trust. By grounding all AI activities in real-world e-business contexts, the course ensured that students perceived their learning as directly applicable to their future careers.

Peer Showcase via Padlet created a social, motivationally rich environment in which students' creative outputs were visible to the entire class community. As students saw their peers producing increasingly sophisticated AI-generated visuals and chatbot interactions, healthy competition emerged naturally, motivating further effort and experimentation. This peer visibility mechanism is consistent with SDT's relatedness need and with research showing that public performance contexts enhance both intrinsic motivation and perceived competence.

Perceived Competence was built progressively through the scaffolded task structure, instructor demonstrations, peer feedback on Padlet, and the visible quality of students' own AI-generated outputs. As students successfully completed each AI task—from generating a product visual to building a fully functional AI chatbot—their confidence in applying AI tools in professional contexts grew measurably. This trajectory from anxiety to competence is consistent with Maslow's self-actualization pathway and with SDT's competence-need fulfilment mechanism.

Table 3 summarizes the full assignment structure deployed within the AI-augmented EBIS, mapping each assessed task to its corresponding RFM Educational Design Model principle, the tools utilized, and the key learning outcomes targeted, illustrating how the three RFM principles were operationalized cohesively across the course rather than in isolation.

Assignment	**RFM Alignment**	**Tools Used**	**Key Outcomes**
Academic paper analysis	Reasonable Challenge: Critical thinking, database research	University library databases, academic journals	Evidence-based argumentation, AI ethics awareness
Product visual design	Fun: Creative, hands-on; Motivation: Perceived usefulness	ChatGPT, DeepAI	Prompt engineering skills, marketing design application
Virtual spokesperson chatbot	Reasonable Challenge: Complex tool; Motivation: Perceived competence	Rochat, ChatGPT (avatar images)	Conversational AI understanding, brand personality design
Padlet showcase participation	Motivation: Peer showcase, relatedness	Padlet	Collaborative learning, healthy competition, feedback skills

Table 3: Assignment Structure and RFM Educational Design Alignment

4. Methodology

4.1 Study Context

The study was conducted in a postgraduate marketing course at a university in Hong Kong during the summer of 2025. The course was designed to expose students to emerging trends and technologies in contemporary marketing practice, with a particular focus on integrating artificial intelligence into e-business marketing workflows.

The course enrolled 19 postgraduate students pursuing full-time master's degrees across business-related disciplines. Participants came from diverse professional backgrounds, representing a range of prior industry experiences in marketing, management, and related fields. Consistent with the broader higher-education context in Hong Kong, most students had limited prior hands-on experience with generative AI tools at the start of the course, making the cohort a representative sample for studying AI competency development at the postgraduate level.

The AI-augmented EBIS deployed in this course integrated five platforms, each fulfilling a distinct functional role within the overall system design. Rather than forming a tightly coupled technical pipeline, the platforms were selected and orchestrated as a complementary toolkit—multiple accessible pathways through which students could achieve the same core learning objectives: creating AI-generated marketing visuals, producing promotional video content, designing conversational AI brand agents, and showcasing their work to peers. This design decision reflected a deliberate pedagogical choice to provide students with flexibility and creative agency rather than prescribing a single fixed workflow.

Platform selection was guided by two explicit design criteria consistent with the Reasonable Challenge principle of the RFM Educational Design Model. First, all platforms were either freely accessible or available through institutional provision, eliminating financial barriers to participation. Second, all platforms offered no-code or low-code interfaces, making them accessible to students without programming backgrounds—an important consideration in a postgraduate business cohort where technical expertise varies widely.

Critically, prior to deployment, the course instructor personally tested each platform on an independent project to assess task difficulty, time requirements, and potential technical barriers. This pre-deployment evaluation ensured that the challenge level presented to students was genuinely reasonable and achievable within the course timeline—reflecting not an assumption but an evidence-informed design decision. For most students, this was their first hands-on experience with these generative AI and chatbot platforms, making the scaffolded introduction and instructor-tested design particularly important for building early confidence and reducing anxiety.

The five platforms fulfilled the following distinct functional roles within the EBIS:

ChatGPT (University Portal): Institutional ChatGPT Plus access provided students with DALL-E image generation capabilities (15 images per day limit), enabling text-to-image creation for marketing visuals. The "Upload Media" function allowed students to provide reference images when struggling to describe desired outputs in text prompts.

DeepAI: Free digital platform offering 100+ stylized image generation models. Positioned as a complementary tool addressing ChatGPT's daily image limit, enabling students to conduct prompt engineering trials before finalizing designs.

PixVerse.ai: Video generation platform converting static AI-generated images into short video clips (5 seconds). Free accounts provided 60 daily credits, sufficient for 1-2 video generations. This tool extended students' generative capabilities from static imagery to animated content.

Rochat: AI-powered chatbot creation platform enabling students to build virtual brand mascots and spokespersons. The platform's low-code interface allowed customization of chatbot appearance (AI-generated

avatars), personality profiles, underlying large language models (LLMs), and voice characteristics without programming knowledge.

Padlet: Digital canvas and collaborative bulletin board enabling students to post AI-generated designs, comment on peers' work, and engage in collective learning. Served as the system's social and showcase component.

These components were orchestrated through structured assignments, in-class demonstrations, tutorial support, and peer interaction protocols, forming an integrated EBIS rather than disconnected tools.

4.2 Data Collection

Quantitative data: Online Learning Environment (OLE) polling conducted in-class using anonymous surveys measuring perceived appeal, usability, usefulness, and ease of access for each AI tool (ChatGPT/DeepAI, Rochat, Padlet). Students rated each dimension on 5-point scales (e.g., "very useful" to "not useful at all").

Qualitative data: (1) Student assignment artifacts including AI-generated imagery, product designs, virtual spokesperson chatbots, and written reflections on academic literature addressing generative AI in marketing[13]; (2) Open-ended survey comments about tool experiences; (3) Peer comments on Padlet showcases.

4.3 Ethical Considerations

Students provided informed consent for anonymized use of their work and survey responses in research. All identifying information was removed from analyzed artifacts.

5. Discussions

5.1 Theoretical Implications

6.1.1 The RFM Educational Design Model as a Design Science Contribution

The RFM Educational Design Model makes a dual theoretical contribution to the EBIS and educational design science literature. First, it emerges not from abstract theorizing but from years of direct pedagogical observation across diverse classroom settings — undergraduate, postgraduate, and distance learning environments — where three recurring patterns became unmistakably clear: when students perceive tasks as excessively challenging, they disengage and give up; when learning feels tedious or boring, participation becomes superficial or performative; and when motivation is absent, students do only the minimum required to pass. These practitioner observations provided the real-world problem that the RFM model was designed to solve — consistent with design science research's requirement that artifacts address genuine, observed problems.

Second, the RFM model advances EBIS design knowledge by providing a theoretically grounded synthesis of five established motivational perspectives — Self-Determination Theory, Expectancy Theory, Attribution Theory, Maslow's Hierarchy of Needs, and Behavioural Learning Theory — translated into actionable design principles for AI-augmented educational environments. While each of these theoretical frameworks has been applied independently in educational research, the RFM model's contribution lies in their deliberate integration into a coherent, practically tested framework with clear operationalization guidance. The three-by-three structure (three main dimensions × three sub-components for Fun and Motivation) offers manageable complexity for instructors designing AI-augmented courses, making theoretical insights directly actionable rather than abstractly aspirational.

Framed within design science research, the RFM Educational Design Model contributes both a design construct (the model itself, with its three principles and associated sub-components) and a set of validated design principles (actionable guidelines for implementing each principle in AI-rich learning contexts) — both of which were empirically tested and demonstrated to produce meaningful outcomes in a real postgraduate marketing course. Critically, this is not a model proposed on paper alone; it is a framework that has been deployed, observed, and evaluated in practice, and recognised through institutional teaching awards for its effectiveness.

6.1.2 Theory-to-Practice Translations: From Motivational Theory to EBIS Design

A key theoretical contribution of the RFM model is its explicit mapping of established motivational theories to concrete EBIS design decisions — a translation that is often missing in educational technology research. Three particularly instructive mappings emerged from this study:

Expectancy Theory → Three-Stage Scaffolding Structure: Expectancy Theory posits that motivation depends on students' perceived likelihood of success. The three-stage scaffolding structure — demonstration, guided experimentation, and independent application — was directly designed to build students' expectation of success before requiring independent performance. By first showing students what was achievable (demonstration), then allowing safe experimentation without grade consequences (guided experimentation), and only then requiring graded independent work (application), the course systematically increased students' confidence that effort would yield meaningful outcomes. The qualitative evidence of students progressing from generic prompts to sophisticated brand-aligned AI outputs confirm that this expectancy-building mechanism worked as theoretically intended.

Attribution Theory → Prompt Engineering Progression: Attribution Theory highlights how students' beliefs about the causes of their success affect their motivation and persistence. As students progressed from ineffective generic prompts to sophisticated, brand-aligned AI-generated content, they could clearly attribute their improvement to their own developing effort and skill — not to luck or the tool itself. This internal attribution of success is a powerful motivational reinforcement, encouraging continued effort and risk-taking in subsequent tasks. The visible contrast between early unsuccessful and later successful outputs made this progression tangible and personally meaningful for students.

Maslow's Hierarchy of Needs → Padlet Peer Showcase: Maslow's framework identifies esteem needs — the need to be recognised, valued, and respected — as a critical motivational driver above basic safety and belonging needs. The Padlet peer showcase mechanism directly addressed students’ esteemed needs by making their creative AI outputs publicly visible and celebrated within the class community. Students' voluntary comments on peers' work and their motivation to produce increasingly impressive designs in response to seeing others' posts reflect the fulfilment of esteem needs predicted by Maslow's framework. This finding suggests that social visibility mechanisms are not merely engagement features but theoretically grounded motivational infrastructure within AI-augmented educational EBIS.

6.2 Managerial Implications

6.2.1 For University Curriculum Designers and Program Directors

The findings of this study offer three main practical implications for institutions seeking to evolve their business curricula to meet the demands of AI-driven e-business environments.

Prioritize high-relevance AI applications as entry points: Student interest data revealed the strongest enthusiasm for AI applications in advertising (29%) and product design (29%), followed by online shopping (24%). These preferences are not arbitrary — advertising and product design represent the AI applications

most immediately relevant to students' anticipated professional roles in marketing and e-business, and also the areas where generative AI tools produce the most visually compelling and immediately gratifying outputs. Curriculum designers should therefore prioritize these high-interest domains as entry points for AI competency development, using them to build confidence and enthusiasm before introducing students to less immediately appealing but strategically important AI applications such as data analytics, moderation, and AI governance. Importantly, the creative skills developed through advertising and product design tasks — prompt engineering, brand alignment, visual communication — transfer readily to other AI application domains, and even to entrepreneurial contexts such as launching new products or creative projects independently.

Adopt the RFM model as a curriculum design framework: The RFM Educational Design Model provides program directors with a structured yet flexible framework for designing AI-augmented courses that are motivationally effective across diverse student cohorts. The model's three principles — Reasonable Challenge, Fun, and Motivation — can be applied at the course level (overall curriculum structure), the module level (individual topic design), and the assignment level (specific task calibration), making it scalable across different program contexts and course sizes.

Invest in tool orchestration, not just tool access: The findings suggest that success in AI-augmented education derives not from access to any single AI platform but from the thoughtful orchestration of complementary tools addressing different competencies — image generation, video creation, conversational AI design, and collaborative showcase. Institutions should therefore move beyond simply providing access to one or two AI tools and instead design integrated AI tool ecosystems that expose students to the range of AI capabilities increasingly deployed in real e-business environments.

6.2.2 For Individual Instructors

Scaffolding is non-negotiable: The evidence clearly demonstrates that direct exposure to AI tools without prior demonstration, guided practice, and progressive challenge produces poor outcomes — generic prompts, superficial engagement, and student anxiety. Individual instructors should invest time in designing structured skill-building pathways, particularly for student cohorts with limited prior AI experience. The three-stage scaffold (demonstrate → guide → apply) provides a replicable template that can be adapted to different tools and course contexts.

Pre-test all tools personally: A critical but often overlooked step in AI-augmented course design is the instructor's own hands-on testing of each platform before deployment. In this study, the course instructor personally tested all five platforms on independent projects to assess difficulty levels, time requirements, and potential technical barriers — ensuring that the challenge presented to students was genuinely reasonable rather than assumed. This pre-deployment evaluation is a practical necessity, not an optional extra.

Embed ethical AI discourse throughout, not as a separate module: Students' spontaneous critical reflections on AI bias, copyright, privacy, consumer trust, and the boundary conditions of virtual versus human influencers demonstrate that ethical AI awareness can emerge organically when assignments require genuine critical engagement with academic literature on generative AI. Instructors should design assignments that prompt students to read, analyze, and form their own positions on AI ethics — rather than delivering ethics as a standalone lecture module disconnected from practical tool use.

Leverage peer learning as a motivational accelerator: Padlet data and student comments confirm that peer examples and feedback accelerated learning more effectively than instructor guidance alone. Instructors should deliberately design social visibility mechanisms — peer showcase platforms, in-class sharing sessions, peer commentary requirements — into their AI-augmented courses, recognizing these not as supplementary activities but as core motivational infrastructure.

6.2.3 Institutional Considerations: Data Privacy and AI Governance

As institutions move beyond classroom AI experimentation toward more sophisticated AI deployments — including AI-powered virtual brand agents that interact with real customers — data privacy and governance

considerations become increasingly critical. In the educational context of this study, privacy risks were limited because chatbot interactions occurred within a controlled classroom environment for skill development. However, institutions and students should be aware that real-world deployment of AI chatbots in customer-facing e-business contexts introduces substantially higher privacy and governance requirements: customer conversation records, data retention policies, and regulatory compliance demand secure, enterprise-grade infrastructure rather than the free educational platforms appropriate for classroom experimentation. Future curriculum iterations should therefore include explicit discussion of the transition from educational AI prototyping to professional AI deployment, equipping students not only with design skills but also with awareness of the governance responsibilities that accompany real-world AI-powered customer interactions.

6. Conclusions

This paper sets out to address a genuine and growing challenge in business education: how do we move beyond talking about artificial intelligence to genuinely teaching with and through it — in ways that are engaging, meaningful, and motivationally sustainable for students?

The answer proposed here is the RFM Educational Design Model — a framework born not in theory but in the honest realities of the classroom. Across years of teaching undergraduate, postgraduate, and distance-learning students, three patterns recurred with quiet persistence: students who found tasks too hard simply gave up; students who found learning tedious or boring merely went through the motions; and students without motivation did only the minimum to pass. These observations — simple, honest, and universal — became the foundation of the RFM model's three principles: Reasonable Challenge, Fun, and Motivation.

The findings are encouraging. Students progressed from generic, ineffective AI prompts to sophisticated, brand-aligned marketing outputs. They built AI-powered virtual brand spokespeople, generated compelling marketing visuals and promotional videos, and engaged in animated, voluntary peer learning through a shared digital showcase. Perhaps most meaningfully, they developed not only technical AI competence but also the kind of spontaneous, nuanced critical thinking about AI ethics, bias, virtual versus human influence, and consumer trust that no tutorial alone can produce. These are the outcomes that matter — not just for passing an assignment, but for navigating a professional world increasingly shaped by AI.

The study contributes to the literature in two ways. Theoretically, it introduces the RFM Educational Design Model as an integrated, theoretically grounded, and empirically validated design framework for AI-augmented educational EBIS — one that has already shown preliminary adaptability across different student populations and learning formats. Practically, it demonstrates that meaningful AI competency development in business education is achievable without expensive infrastructure, programming expertise, or institutional-scale investment. Free and low-cost platforms, thoughtfully scaffolded and motivationally designed, are sufficient to produce genuine learning outcomes — a finding with important implications for institutions at all stages of their AI education journey. The approach has also been independently recognized for its effectiveness in fostering student engagement and learning outcomes, providing external validation beyond the authors' own observations.

The RFM model is not offered here as a finished answer but as an open invitation. An invitation to educators who have observed the same three patterns in their own classrooms — and who believe, as the authors do, that students thrive when they are challenged appropriately, when learning is genuinely enjoyable, and when they feel motivated to succeed. An invitation to institutions ready to evolve their curricula from passive AI literacy to active AI competency. And an invitation to researchers who will extend, challenge, and refine these ideas across disciplines, cultures, and learning contexts that this study could not reach.

References

[1] Amalia, R., Paradita, J., Aini, R. Q., Sudrajat, Mirza, A. D., & Prayoga, R. S. (2024). Transforming business with AI: Impacts, challenges and opportunities. In Proceedings of the EAI Conference on AI and Digital Transformation. EAI. https://eudl.eu/pdf/10.4108/eai.4-9-2024.2353765

[2] Ashkani, M., Dana, L.-P., Rashidi, A., Shafaei, F., & Salamzadeh, A. (2026). Drivers and sustainable performance outcomes of AI adoption intention: A multi-theoretical analysis in the entrepreneurial ecosystem. Sustainability, 18(3), 1417. https://doi.org/10.3390/su18031417

[3] Bashir, T., Tan, Z., Sadiq, B., & Naseem, A. (2024). How AI competencies can make B2B marketing smarter: strategies to boost customer lifetime value. Frontiers in Artificial Intelligence, 7, 1451228. https://doi.org/10.3389/frai.2024.1451228

[4] Chen, L. (2022). Current and future artificial intelligence (AI) curriculum in business school: A text mining analysis. Journal of Information Systems Education, 33(4), 416–426. https://aisel.aisnet.org/jise/vol33/iss4/8/

[5] Chow, C. S. K., Zhan, G., Wang, H., & He, M. (2023). Artificial intelligence (AI) adoption: An extended compensatory level of acceptance. Journal of Electronic Commerce Research, 24(1), 111–132. http://www.jecr.org/sites/default/files/2023vol24no1_Paper6.pdf

[6] Dar, S. A., & Sakthivel, P. (2022). Maslow's hierarchy of needs is still relevant in the 21st century. Journal of Learning and Educational Policy, 2(5), 1–9. https://doi.org/10.55529/jlep25.1.9

[7] Estevez, M., Ballestar, M. T., & Sainz, J. (2025). Market research and knowledge using Generative AI: the power of Large Language Models. Journal of Innovation & Knowledge, 10(5), 100796. https://doi.org/10.1016/j.jik.2025.100796

[8] Fang, X., Ng, D. T. K., Leung, J. K. L., & Xu, H. (2024). The applications of the ARCS model in instructional design, theoretical framework, and measurement tool: A systematic review of empirical studies. Interactive Learning Environments, 32(10), 5919-5946. https://doi.org/10.1080/10494820.2023.2240867

[9] Frei-Landau, R., & Levin, O. (2023). Simulation-based learning in teacher education: Using Maslow's hierarchy of needs to conceptualize instructors' needs. Frontiers in Psychology, 14, 1149576. https://doi.org/10.3389/fpsyg.2023.1149576

[10] Gonzalez, R. A., Wareham, J., & Serida, J. (2012). Validation and design science research in information systems. In K. Peffers, M. Rothenberger, & B. Kuechler (Eds.), Design science research in information systems (pp. 1–15). Springer. https://pure.rug.nl/ws/portalfiles/portal/225614850/Validation_and_Design_Science_Research_in_Information_Systems.pdf

[11] Grewal, D., Satornino, C. B., Davenport, T., & Guha, A. (2025). How generative AI is shaping the future of marketing. Journal of the Academy of Marketing Science, 53(3), 702-722. https://doi.org/10.1007/s11747-024-01064-3

[12] Hamari, J., Shernoff, D. J., Rowe, E., Coller, B., Asbell-Clarke, J., & Edwards, T. (2016). Challenging games help students learn: An empirical study on engagement, flow, and immersion in game-based learning. Computers in Human Behavior, 54, 170–179. https://doi.org/10.1016/j.chb.2015.07.045

[13] Hartmann, J., Exner, Y., & Domdey, S. (2025). The power of generative marketing: Can generative AI create superhuman visual marketing content? International Journal of Research in Marketing, 42(1), 13–31. https://doi.org/10.1016/j.ijresmar.2024.09.002

[14] Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75–105. https://www.jstor.org/stable/25148625

[15] Howe, D., & Soderberg, A. (2025). The court of theories: a debate-style exercise for teaching motivation theories. Organization Management Journal, 22(3-4), 203-213.

https://www.emerald.com/omj/article/22/3-4/203/1269102/The-court-of-theories-a-debate-style-exercise-for

[16] Kasap, S., & Ünsal, F. (2021). Attribution theory and language learning. Research in Language and Education: An International Journal, 2, 111–118.

[17] Kshetri, N., Dwivedi, Y. K., Davenport, T. H., & Panteli, N. (2024). Generative artificial intelligence in marketing: Applications, opportunities, challenges, and research agenda. International journal of information management, 75, 102716. https://doi.org/10.1016/j.ijinfomgt.2023.102716

[18] Landers, R. N. (2014). Developing a theory of gamified learning: Linking serious games and gamification of learning. Simulation & Gaming, 45(6), 752–768. https://doi.org/10.1177/1046878114563660

[19] Luarn, P., Chen, C. C., & Chiu, Y. P. (2023). Enhancing intrinsic learning motivation through gamification: A self-determination theory perspective. The International Journal of Information and Learning Technology, 40(5), 413–424. https://doi.org/10.1108/IJILT-07-2022-0145

[20] Mallary, K. J., Moore, E. J., & McClain, A. L. (2025). Artificial Intelligence and Universal Design for Learning: Transforming Teaching and Learning in Adult and Continuing Higher Education. New Directions for Adult and Continuing Education, 2025(188), 39-47.

[21] Malle, B. F. (2022). Attribution theories: How people make sense of behavior. In J. Olson (Ed.), Theories in social psychology (2nd ed., pp. 93–120). Wiley. https://research.clps.brown.edu/SocCogSci/Publications/Pubs/Malle%20(2022)%20Attribution%20theories.pdf

[22] Malone, T. W., & Lepper, M. R. (2021). Making learning fun: A taxonomy of intrinsic motivations for learning. In Aptitude, learning, and instruction (pp. 223-254). Routledge.

[22] Maslow, A. H. (1943). A theory of human motivation. Psychological Review, 50(4), 370–396. https://doi.org/10.1037/h0054346

[23] Memmert, L., Soroko, D., & Bittner, E. (2025). From effort reduction to effort management: An expectancy theory perspective on professionals' work practices with generative AI. Business & Information Systems Engineering. Advance online publication. https://link.springer.com/content/pdf/10.1007/s12599-025-00960-4.pdf

[24] Mikalef, P., Islam, N., Parida, V., Singh, H., & Altwaijry, N. (2023). Artificial intelligence (AI) competencies for organizational performance: A B2B marketing capabilities perspective. Journal of Business Research, 164, 114328. https://doi.org/10.1016/j.jbusres.2023.113998

[25] Motevalli, S., Perveen, A., & Michael, M. T. A. (2020). Motivating students to learn: An overview of literature in educational psychology. International Journal of Academic Research in Progressive Education and Development, 9(3), 63–74. https://doi.org/10.6007/ijarped/v9-i3/7779

[26] Prasanna, A., & Kushwaha, B. P. (2025). Generative AI in marketing: foundations, trends, and future research propositions. Human Behavior and Emerging Technologies, 2025(1), 5542513. https://doi.org/10.1155/hbe2/5542513

[27] Royal, C. (2026). Integrating vibe coding and Flow Theory: A student-centered model for AI-augmented coding education. Journalism & Mass Communication Educator. Advance online publication. https://doi.org/10.1177/10776958251407389

[28] Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. American Psychologist, 55(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

[29] Ryan, R. M., Deci, E. L., Vansteenkiste, M., & Soenens, B. (2021). Building a science of motivated persons: Self-determination theory's empirical approach to human experience and the regulation of behavior. Motivation Science, 7(2), 97–110. https://doi.org/10.1037/mot0000194

[30] Sailer, M., & Homner, L. (2020). The gamification of learning: A meta-analysis. Educational Psychology Review, 32(1), 77–112. https://doi.org/10.1007/s10648-019-09498-w

[31] Thakur, C. (2025). Exploring the influence of generative AI on consumer experiences and behavioral responses: A systematic literature review. Journal of Marketing Theory and Practice, 33(2), 145–167. https://www.abacademies.org/articles/exploring-the-influence-of-generative-ai-on-consumer-experiences-and-behavioral-responses--a-systematic-literature-revie.pdf

[32] Weiner, B. (1985). An attributional theory of achievement motivation and emotion. Psychological Review, 92(4), 548–573. https://doi.org/10.1037/0033-295X.92.4.548

[33] Yaprak, B., Ercan, S., Coşan, B., & Ecevit, M. Z. (2025). Curriculum–skill gap in the AI era: Assessing alignment in communication-related programs. Journal of Intelligence and Responsible Communication, 6(4), 171. https://doi.org/10.3390/journalmedia6040171